Table 9. **Comparison between previous state-of-the-art single-model-based methods and ours on the SC-OOD CIFAR10 benchmarks.** The results are reported for models with ResNet-18 backbone. Best score in **bold**, second best underlined.

Method	Additional Data	FPR95 ↓	AUROC ↑	AUPR ↑
ODIN	✖	52.0	82.0	85.1
EBO	✖	50.0	83.8	85.1
OE	✔	50.5	88.9	87.8
MCD	✔	73.0	83.9	80.5
UDG	✖	55.6	90.7	88.3
UDG	✔	**36.2**	**93.8**	**92.6**
Split-Ensemble (ours)	✖	45.5	91.1	89.9

Table 10. **Comparison between previous state-of-the-art ensemble-based methods and ours on the SC-OOD CIFAR10 benchmarks.** The results are reported for models with ResNet-18 backbone. Best score in **bold**, second best underlined.

Method	FLOPs	FPR95 ↓	AUROC ↑	AUPR ↑
Naive Ensemble	4x	42.3	90.4	90.6
MC-Dropout	4x	54.9	88.7	88.0
MIMO	4x	73.7	83.5	80.9
MaskEnsemble	4x	53.2	87.7	87.9
BatchEnsemble	4x	50.4	89.2	88.6
FilmEnsemble	4x	**42.6**	**91.5**	**91.3**
Split-Ensemble (ours)	1x	45.5	91.1	89.9

Table 11. **Accuracy (↑)/expected calibration error (↓) on the corrupted CIFAR-10-C dataset**. Best score for each metric in **bold**, second-best underlined. We implement all baselines using default hyperparameters. All accuracies/ECEs are given in percentage with ResNet-18 as backbone.

Method	FLOPs	Clean	Severity 1	Severity 2	Severity 3	Severity 4	Severity 5
Naive Ensemble	4x	**95.7** / 3.9	**91.4** / 7.7	**85.1** / 13.6	78.6 / 19.7	71.9 / 26.0	59.4 / 37.7
MC-Dropout	4x	92.4 / 3.5	88.3 / 5.7	82.7 / 9.5	77.0 / 13.6	70.9 / 18.2	59.6 / 26.2
MIMO	4x	84.4 / **1.1**	79.0 / **3.4**	72.0 / **6.9**	65.7 / **10.8**	59.2 / **15.1**	49.8 / **21.4**
MaskEnsemble	4x	94.2 / 3.3	89.3 / 6.9	83.6 / 10.9	77.6 / 15.4	70.9 / 20.2	58.9 / 29.9
BatchEnsemble	4x	93.8 / 3.1	88.8 / 5.7	82.7 / 10.1	76.1 / 15.4	69.4 / 20.2	57.6 / 29.1
FilmEnsemble	4x	93.8 / 3.0	89.1 / 6.5	84.1 / 10.5	77.7 / 13.9	71.5 / 20.2	59.2 / 27.2
Split-Ensemble (Ours)	1x	95.5 / 4.1	90.7 / 7.0	84.8 / 10.5	**79.2** / 13.7	**73.2** / 16.7	**61.7** / 22.7

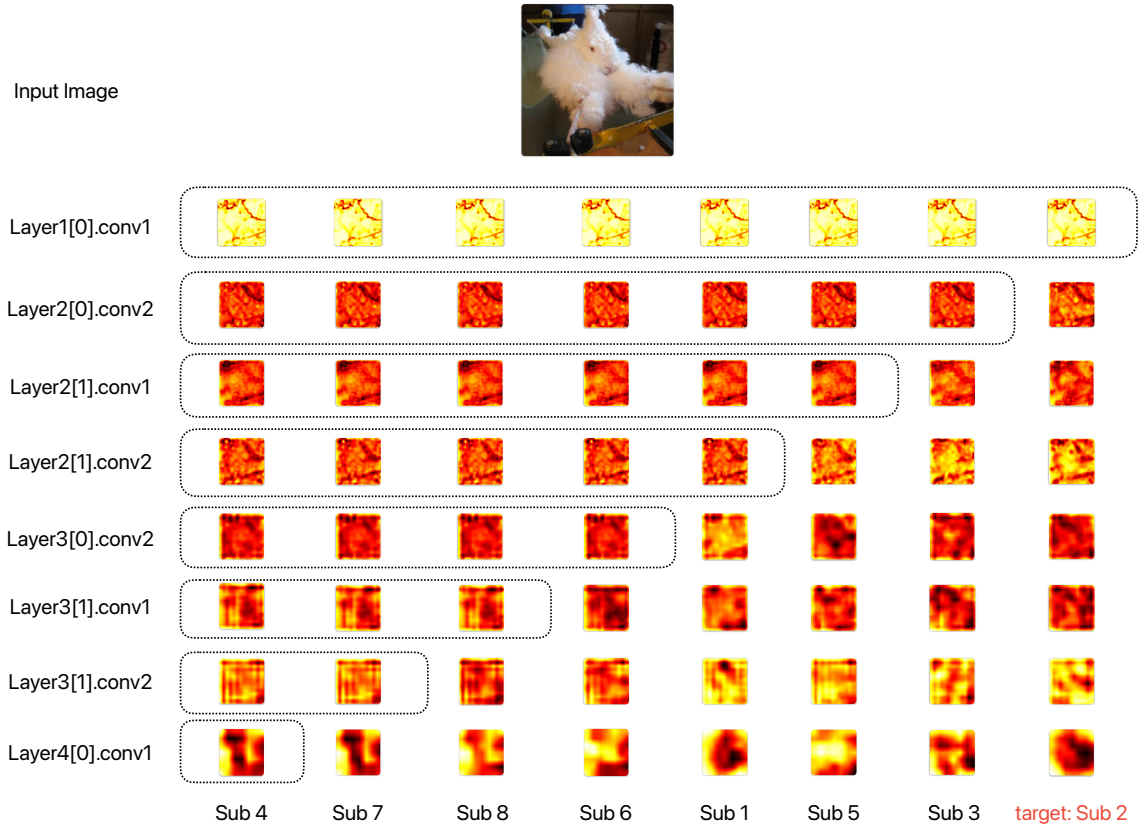

Figure 5. **Visualization of Split-Ensemble's learned features using Score-CAM.** The number of splits is set to 8 and the model is trained on ImageNet1K with ResNet-18 as backbone. The feature maps within the dashed lines across the layers indicate shared representations. The input image's class is 'Angora', targeted by submodel 2.

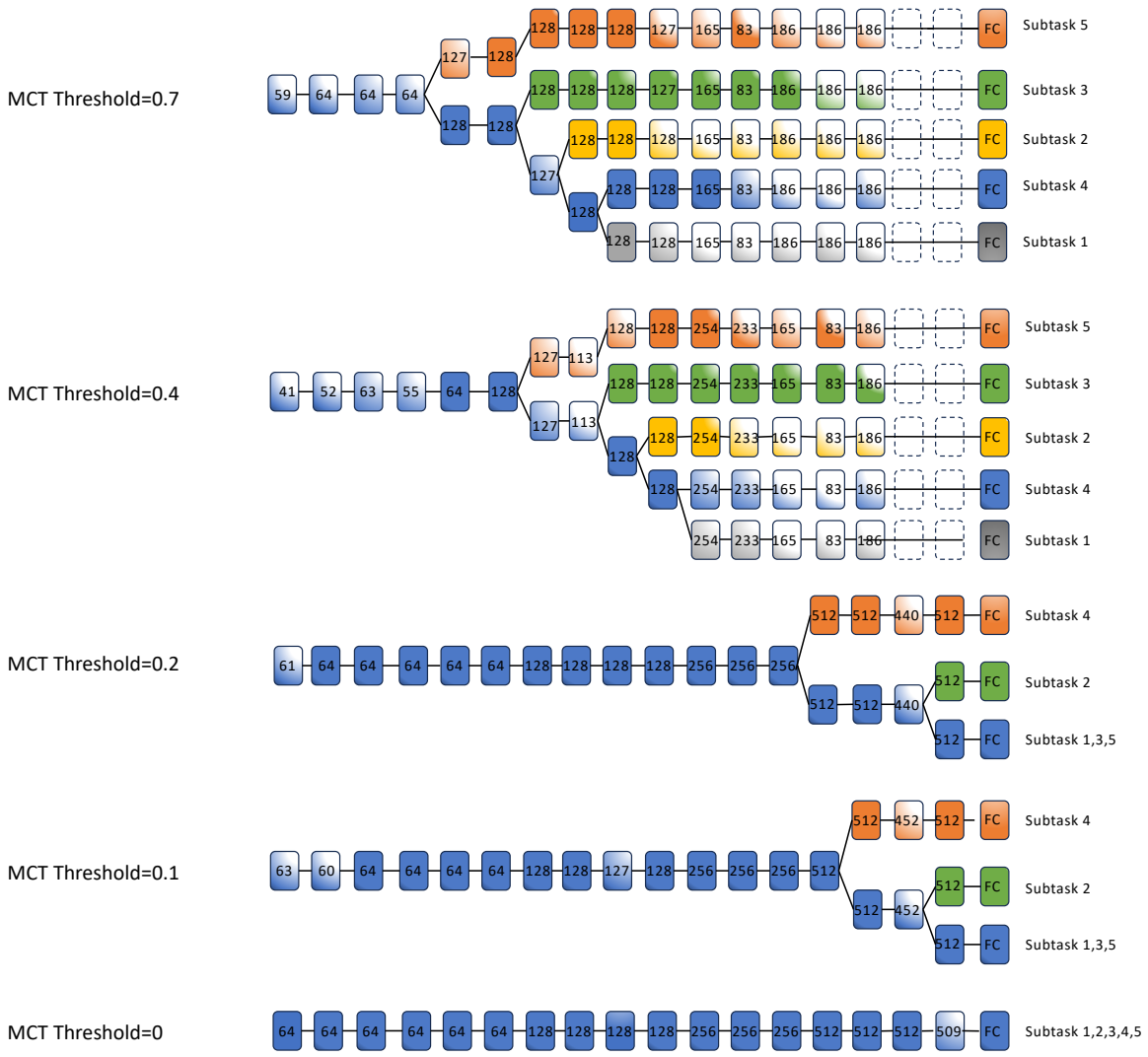

Figure 6. **Visualization of Split-Ensemble architectures under different MCT threshold.** The number of splits is set to 5. Number in each block denotes the number of filters in the layer.